

Review Three for the Final Exam

1 Final Exam

- on Friday 7 August from 2:00pm till 3:40pm
- focus of this review

2 Numbers and Expressions

- first steps
- polynomials and expressions

MCS 320 Lecture 41
Introduction to Symbolic Computation
Jan Verschelde, 5 August 2026

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- You must sign up to the exam before noon on Thursday 6 August by sending an email that you commit to taking the exam.
- Without signing up, you will not receive the questions, and your final grade will be an F.
- You must solve the questions on the exam *individually*.
- The exam will have multiple versions of the questions.
- The answers must be uploaded into gradescope
 - ▶ in one single Jupyter notebook,
 - ▶ before 3:40pm on Friday 7 August 2026.

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Focus of This Review

This is the third review, with a focus on the first two parts.

- Look at the review of the first midterm exam.
- The answers to the first midterm exam are posted.
Even if you took the exam, look at the other version.
- Consider the answers to the quizzes.
- Review the homework problems.

Make sure you understand your mistakes.

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First Steps

The first 10 lectures (5 sessions) cover the first part.

- SageMath in a Jupyter notebook, code and markdown cells
- symbols, exact and floating-point approximations
- complex numbers and algebraic numbers, finite fields
- sharing objects with references, functions to store data
- binary expression trees for fast callable objects
- apply vectorization for speed

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Polynomials and Expressions

Lectures 11 to 15 (3 sessions) cover the second part.

- polynomials, quotients, multivariate polynomials, expressions
- numeric, exact, symbolic, and symbolic-numeric factorization (symbolic-numeric factorization is covered at the end of part 3)
- expression trees are not binary
- manipulating expressions by substitution, expansion, factorization
- normal forms to solve the equality problem